



Public Economics

Lecture 8 · Public Goods

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- What makes a good “public” in the economist’s sense — and why does the market systematically undersupply it?
- If everyone has an incentive to free-ride, why does *any* private giving happen at all?
- Does government provision simply replace private giving one-for-one, or can the two coexist?

Based on

Saez, E., *Econ 131* lecture notes (UC Berkeley); Gruber, J. *Public Finance and Public Policy*, Ch. 7.

7.1 Optimal Provision of Public Goods

7.2 Private Provision of Public Goods

7.3 Crowding Out and Government Provision

7.1 Optimal Provision of Public Goods

Pure public good

Perfectly **non-rival in consumption** and **non-excludable**.

- **Non-rival:** one person's consumption doesn't reduce anyone else's opportunity to consume it.
- **Non-excludable:** nobody can be stopped from consuming it.

Impure public good

Satisfies the two conditions only *partially* — most real-world “public goods” are actually this.

The economist's definition is narrow: it says nothing about whether a good *should* be universally provided. Education and health care are not public goods in this technical sense — they're rival and (at least partly) excludable — even though most societies choose to provide them broadly.

Two goods: ice cream (ic) and cookies (c , the numéraire, $P_c = 1$). Two individuals, B and J , face the same market price P_{ic} but may consume different quantities.

- $MRS_{ic,c} = MU_{ic}/MU_c$ = cookies a consumer will give up for one more ice cream.
- **Individual optimum:** $MRS_{ic,c}^B = MRS_{ic,c}^J = P_{ic}$.
- **Supply-side equilibrium:** $MC_{ic} = P_{ic}$.

$$\Rightarrow MRS_{ic,c}^B = MRS_{ic,c}^J = MC_{ic}$$

Each person separately equates their own MRS to the (common) marginal cost — because one more ice cream for B doesn't affect J at all.

Now replace ice cream with a public good — missiles (m) — consumed by *both* B and J **simultaneously**.

- $MRS_{m,c}^B$ = cookies B will give up for one more missile; $MRS_{m,c}^J$ likewise for J .
- Since both enjoy the same missile, society is willing to give up $MRS_{m,c}^B + MRS_{m,c}^J$ cookies for it.

Samuelson Rule (Samuelson, 1954)

$$MRS_{m,c}^B + MRS_{m,c}^J = MC_m$$

Efficiency requires marginal cost to equal the **sum** of everyone's MRS — not each person's MRS separately, as with a private good.

7.2 Private Provision of Public Goods

Left to itself, each individual i contributes only up to the point where *their own* $MRS_{m,c}^i = MC_m$ — ignoring the benefit to everyone else.

$$\sum_i MRS_{m,c}^i > MC_m \Rightarrow \text{too little of the public good is provided.}$$

Free-rider problem

When an investment has a personal cost but a common benefit, selfish individuals will underinvest.

Conceptually: private provision of a public good creates a **positive externality** on everyone else — and, exactly as with any positive externality, the market undersupplies it.

Providing public goods is the oldest function of government. The largest public goods are supplied almost entirely by the state:

1. National defense.
2. Police and justice.
3. Transportation, energy, and sanitation infrastructure.
4. Increasingly: environmental protection.

Basic science is typically government-funded, while applied R&D is mostly private; a technology becomes a genuine public good once it falls into the public domain.

Two identical individuals share a private good X (cookies) and jointly fund a public good F (fireworks), $F = F_1 + F_2$.

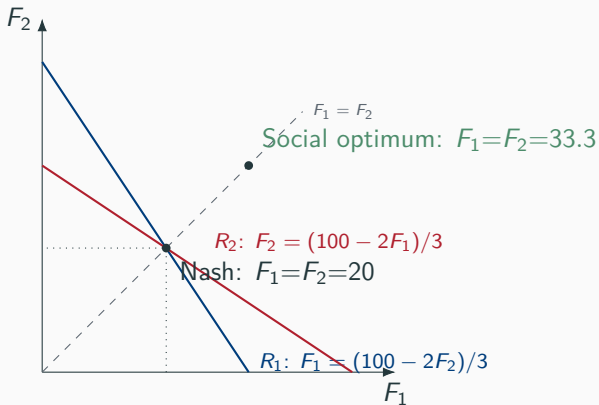
$$U_i = 2 \log(X_i) + \log(F_1 + F_2), \quad X_i + F_i = 100$$

Individual 1 maximizes $2 \log(100 - F_1) + \log(F_1 + F_2)$ taking F_2 as given:

$$-\frac{2}{100 - F_1} + \frac{1}{F_1 + F_2} = 0 \quad \Rightarrow \quad F_1 = \frac{100 - 2F_2}{3}$$

This is individual 1's **reaction curve**: F_1 falls as F_2 rises — the free-rider effect in action. By symmetry, $F_2 = (100 - 2F_1)/3$.

Diagram: Reaction Curves and the Nash Equilibrium



The two reaction curves cross at the Nash equilibrium, well *below* the socially optimal contribution on the $F_1 = F_2$ line.

At the Nash equilibrium, both reaction curves hold simultaneously:

$$F_1 + F_2 = \frac{200 - 2(F_1 + F_2)}{3} \Rightarrow F = F_1 + F_2 = \frac{200}{5} = 40 \Rightarrow F_1 = F_2 = 20$$

The social optimum requires $\sum_i MRS^i = MC = 1$, with $MRS_{F,X}^i = \frac{X_i}{2F}$:

$$\sum_i MRS^i = \frac{200 - F}{2F} = 1 \Rightarrow F = \frac{200}{3} = 66.6 > 40$$

Public goods are under-provided by the market

Free-riding cuts total fireworks nearly in half relative to the efficient level.

Setup

You're in a group of 5. You have 10 tokens. 1 token kept privately = \$1 for you. 1 token given to the group = \$0.50 to *each* of the 5 members (including you) — \$2.50 total.

How would you split your 10 tokens?

- A. All 10 to the private good.
- B. All 10 to the public good.
- C. Split them half and half.

Nash equilibrium (selfish play): keep everything privately. Socially optimal: contribute everything — \$1 kept is worth less to the group than \$2.50 shared.

Marwell & Ames (1981): 10 repetitions of the game above (group of 5, 10 tokens each; private return \$1, public return \$0.50 to each of 5).

- **Nash prediction:** contribute nothing.
- **Social optimum:** contribute everything.
- **What actually happens:** subjects contribute roughly **50%** to the public good.

Isaac, McCue & Plott (1985): contributions **decline** the more the game is repeated — people start out willing to cooperate, then get upset and retaliate once they see others taking advantage.

Source: Marwell & Ames (1981); Isaac, McCue & Plott (1985).

In the standard model, selfish individuals play Nash and free-ride. Yet humans plainly cooperate constantly — in families, workplaces, communities, nations.

- Cooperation appears to be **innate**, sustained by a sense of fairness and a willingness to **punish non-cooperators** even at a cost to the punisher (“altruistic punishment”).
- Likely an evolutionary adaptation.
- Fehr & Schmidt (1999) formalize “fairness preferences” in a large lab literature.

These social motives are still only partially integrated into mainstream economic models — a real limitation, especially for public economics.

7.3 Crowding Out and Government Provision

Suppose government now **forces** each individual to contribute 5, so $F = F_1 + F_2 + 10$, where F_i is now *voluntary* top-up:

$$U_i = 2 \log(X_i) + \log(F_1 + F_2 + 10), \quad X_i + F_i = 95$$

Solving gives $F_1 = F_2 = 15$ — **one-for-one crowd-out** of the forced contribution.

- **Why?** Let $F'_i = F_i + 5$. Then $U_i = 2 \log(X_i) + \log(F'_1 + F'_2)$ with $X_i + F'_i = 100$ — *identical* to the original (no-government) problem, so F'_i must be the same: $F'_i = 20 \Rightarrow F_i = 15$.
- Crowd-out stops being one-for-one only once voluntary contributions hit **zero** — you can't “un-contribute” a negative amount.

Crowd-out

The reduction in private contributions caused by an increase in government provision of the same public good.

Two strands of evidence:

1. **Field evidence** (observational studies).
2. **Lab and field experiments.**

Lab experiments (Andreoni, 1993) find crowd-out is real but **imperfect** — well below the one-for-one prediction of the simple model. Lab settings likely miss important real-world motives for giving: warm glow, prestige, and active solicitation by fundraisers.

Charitable giving is one form of private public-good provision.

- In the US, giving totals about **1.5% of national income** — large in absolute terms, but far smaller than the gap in government spending between the US ($\approx 30\%$ of national income) and the EU ($\approx 45\%$).
- Funds flow mainly to: religious organizations, education, human services, health, the arts, and other causes (environment, etc.).
- Governments actively **encourage** it: charitable gifts are typically deductible from taxable income.

1. **Warm glow** — e.g., your name on a building.
2. **Reciprocity** — e.g., alumni giving back to their university.
3. **Social pressure** — e.g., giving at church, in front of others.
4. **Altruism** — e.g., poverty relief for its own sake.

None of these motives are captured by the basic free-rider model above — which is exactly why charities run large fundraising operations built around social and psychological cues, not just the pure economics of public-good provision.

Government spending can crowd out private donations through **two** channels: reduced willingness to give, *and* reduced fundraising effort.

- Panel data on arts and social-service organizations' tax returns, followed over time.
- **Finding:** a \$1,000 increase in a government grant leads to about \$250 less private fundraising.

Most of the crowd-out operates through the **fundraising** channel: less government funding means organizations fundraise harder (e.g., federal funding cuts to universities tend to boost their own fundraising drives).

Crowding in

Kotchen & Wagner (2023): when the US spends more on national parks, **volunteer hours rise too** — government spending can crowd private effort *in*, not just out.

Falk (2007): 10,000 Christmas solicitation letters (funding schools for street children in Bangladesh) sent to Swiss donors, randomized into three groups.

Group		Gift included	Response rate
Control		None	12%
Treatment 1	Small (1 postcard + envelope)		14%
Treatment 2	Large (4 postcards + envelopes)		21%

A larger upfront gift substantially raised giving — consistent with reciprocity, and effective even relative to its cost.

Source: Falk (2007), *Econometrica* 75(5).

DellaVigna, List & Malmendier (2012): a randomized door-to-door fundraiser experiment.

- **Control:** no advance warning of the visit.
- **Treatment 1:** a doorknob flyer announces the exact visit time (so residents can seek out — or avoid — the fundraiser).
- **Treatment 2:** same flyer, plus a “Do not disturb” checkbox.

Results (vs. control): Treatment 1 — 9–25% less likely to open the door, unconditional giving unchanged. Treatment 2 — many opt out, and unconditional giving falls 28–42%. Social pressure is a real determinant of door-to-door giving — and the fundraiser’s visit itself *lowers* the utility of potential donors.

Beyond taxes and subsidies, policy can try to shift **social norms** directly — “social prices.”

- Make a positive-externality behavior a status symbol: electric cars are increasingly admired, while large gas-guzzling SUVs are increasingly frowned upon (Butera et al., 2022).
- Social-price elasticities can be large: Gerber, Green & Larimer (2008) show that a letter using social pressure (“your neighbors will see whether you voted”) substantially raises voter turnout.

Should social pricing be layered *on top of* standard corrective taxes or tradable permits?

- **Shaming SUV drivers** is generally **inefficient** relative to a gas tax: it destroys welfare without raising any revenue. It can still be worthwhile when a fine or tax is hard to enforce at all — e.g., social disapproval of littering, where fines are difficult to levy.
- **Praising efficient-car owners**, by contrast, is **efficient** relative to a gas tax: it adds to welfare, since driving the energy-efficient car becomes more enjoyable in itself.

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Thank you!

Questions and discussion